

Norwood Systems: Reinventing voicemail for the digital age with powerful, versatile Speech-to-Text

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ABOUT NORWOOD

norwood
connect globally, locally™

Norwood System
innovative AI-driv
voicemail solutio
expanded global
Anthos and Goog
API.

About Norwood Systems

Norwood Systems is an Australian telecommunications company on a mission to reboot voicemail for the digital era with World Voicemail, a Transcribed Visual Voicemail Telco Platform. The white-label solution provides global telcos multiple AI features, including automatic voicemail transcription in multiple languages, within a user-friendly app. World Voicemail was named a finalist in Australia's 2019 National iAwards.

Google Cloud results

- Enabled development of trilingual voicemail solution with robust Speech-to-Text feature set
- Supported global expansion to five markets by enabling compatibility with any telco architecture using Anthos
- Enhanced customer satisfaction by enabling seamless app updates with Firebase Remote Config

Industries: Telecommunications

Location: Australia

Enabled to five in 18 languages

Google Cloud (<https://cloud.google.com/>)

Cloud Speech to Text
(<https://cloud.google.com/speech/>)

Contact us (</contact>)
Anthos (<https://cloud.google.com/anthos/>)

TensorFlow (<https://www.tensorflow.org/>)

In an age in which an estimated 2.27 trillion

(<https://venturebeat.com/2013/05/08/five-reasons-you-should-be-using-sms-based-marketing/>)

text messages are sent per year in the United States alone, the conventional wisdom is that voicemail is going the way.

(<https://accesstech.biz/the-way-we-connect-has-changed-is-voicemail-dead/>)

of faxes and floppy disks. Australian telco innovator

Norwood Systems (<https://norwoodsystems.com/>),

however, believes that innovation is born of bucking conventional wisdom. And it is proving that voicemail has a vibrant future through its products and solutions.

From Australia to Sweden and the US, Norwood is breathing new AI-powered life into neglected technology with an innovative solution called World Voicemail

(<https://norwoodsystems.com/guide/docs/world-voicemail/>)

, using Speech-to-Text (/speech-to-text).

Imagine being an electrician in bustling multi-ethnic Sydney serving customers in both the Chinese and Thai communities. You get more than 50 voicemails per day from people who all feel they have an emergency that can't be left to languish in a website inbox.

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—Mans Shapshak, Vice
President Global
Professional Services and
Analytics, Norwood
Systems

Norwood's AI solution allows the voicemail power-user to quickly parse and prioritize the most important jobs, save back-burner calls from friends who want to meet for a beer, and kill spam messages.

To reboot voicemail for the digital age, Norwood needed the most powerful and complete voice-to-text available to enable instant text conversion, with consistent quality, in the widest range of languages possible. It also needed the agility to serve any telco with its white-label solution, for seamless connection whether the client is on-premises, in a private cloud, or a hybrid or multicloud environment.

Norwood found the ideal combination in Speech-to-Text and Anthos (/anthos). Anthos enables Norwood to achieve seamless, secure compatibility with any client. Along with Speech-to-Text, they enable World Voicemail to achieve both a global language footprint and universal compatibility with any architecture. Crucially, out-of-box AI tools mean Norwood can focus on building cool features instead of underlying technology.

"Cloud Speech-to-Text is simply the most complete voice-to-text API, enabling us to offer more functionality, with maximum speed. The best thing is we don't have to build anything from scratch," says Mans Shapshak, Vice President, Global Professional Services and Analytics at Norwood Systems. "With that agility, Anthos then enables us to integrate with as many telcos as needed, by providing a consistent platform for any app deployment."

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—Aaron D'Cruz, Product
Marketing Specialist,
Norwood Systems

Superior Speech-to-Text and universal compatibility for a versatile voicemail solution

Norwood World Voicemail enables countless telco customers in Europe, North America, and the Asia-Pacific, assisted by Google AI tools. In particular, Speech-to-Text automatically recognizes and transcribes more than 150 languages, from Swedish to Swahili, with consistent accuracy across the board. Superior language range enables Norwood to offer users a unique functionality: selecting up to three languages in the app that are automatically rendered into text, whenever a voicemail lands.

Since launching World Voicemail with Speech-to-Text in July 2018, Norwood has expanded to five markets globally, in 18 languages.

Referring to the Sydney electrician example, it would be a common occurrence to see messages in Thai, English, and Chinese text scroll one after another. But every once in a while, the phone buzzes with an urgent app notification from a customer who has a real emergency requiring immediate attention.

In scenarios like this, [TensorFlow](https://www.tensorflow.org/) (<https://www.tensorflow.org/>) supports Norwood's proprietary AI models that intelligently analyze voicemail vocabulary and grammar for both urgency and spam. It means genuine emergencies are alerted with a push notification, even as junk messages are flushed out. Features such as word search, archiving, and playback speed adjustment enhance the convenience.

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transcription
solution using
Speech-to-Text.
And don't worry
about
compatibility
with your cloud
provider. We can
hook everything
up seamlessly."

*—Paul Ostergaard, CEO,
Norwood Systems*

"We found Speech-to-Text to have the most
complete feature set including speed, consistent
accuracy, and breadth of languages," says Aaron

D'Cruz, the company's product marketing specialist.

"Since we're all about creating the most complete voicemail solution, Google was the natural choice for us. That enabled us to quickly expand around the world."

Of course, to achieve such global expansion, World Voicemail must function on whatever computing architecture its telco client is built on. In particular, Anthos is ideally suited to connect World Voicemail's container-based software to any telco architecture (since Anthos itself is a Kubernetes-driven system).

"At Norwood, we're focused on whatever our customers need. White labeling is the crux of our business," says Paul Ostergaard, Norwood CEO.

"Google Cloud and Anthos provide us a compelling argument to win a telco's business. We can say we have this great transcription solution using Speech-to-Text. And don't worry about compatibility with your cloud provider. We can hook everything up seamlessly."

"The best part about Anthos is it's so simple," he adds. "We just plug it in and it works immediately."



Delighting customers with frictionless application updates using Firebase Remote Config

The job of delighting global telcos (and their customers) only begins with architecture integration. To keep them engaged and loyal, it's necessary to continuously improve the solution, making the process so natural that people barely notice it's happening.

This requires a continuous stream of app updates, with zero downtime and frictionless user experience. Norwood executes seamless global system upgrades using Firebase Remote Config (<https://firebase.google.com/products/remote-config>). It's a cloud service that lets programmers change the behavior and appearance of their app, without requiring users to download an app update.

"It's invaluable for us to be able to remotely update our solution without submitting a new build to the app stores," says D'Cruz, "which would cause end users a lot of hassle."

Using Firebase Remote Config, Norwood has been able to rapidly update app configurations five times per month, releasing 50% fewer app updates. This allows the company to run experiments over a contained cohort.

Norwood plans to continue building the future of voicemail with Google AI tools. One work-in-progress with the Google team is automatic transcription of different languages within a single voice message, in cases where the caller switches from tongue to tongue.

Norwood is exploring adoption of more Google AI solutions both for app performance and the next chapter in its AI modeling, opening an unlimited horizon of innovation.

"Our road map includes engagement with Google AI solutions for platform advances we're only just beginning to conceptualize," says Shapshak. "There are many exciting developments ahead."